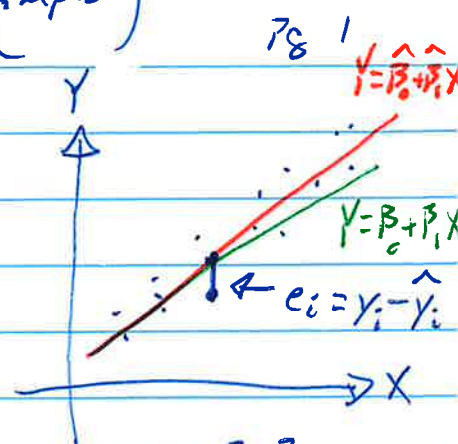


(Multiple) Multinomial Linear Regression (extension of simple LR)

Recap: $Y = f(X) + \epsilon = (\beta_0 + \beta_1 X) + \epsilon$



$$Y \approx \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

①

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \\ &= \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} \end{aligned}$$

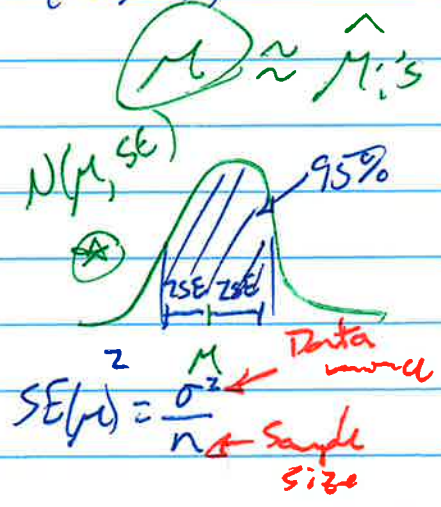
via minimizing $\sum e_i^2$

$$\begin{aligned} &= \sum (y_i - \hat{y}_i)^2 \\ &= \sum (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 \end{aligned}$$

Inference $\rightarrow$ How to quantify certainty around $(\hat{\beta}_0, \hat{\beta}_1)$?

95% CI: $\hat{\beta}_0 \pm 2SE(\hat{\beta}_0)$ formulas!

$\hat{\beta}_1 \pm 2SE(\hat{\beta}_1)$



HIT: $H_0: X \text{ \& } Y \text{ not assoc. } (\beta_1 = 0)$ *assume true*

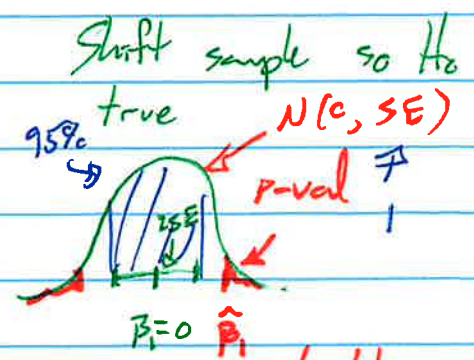
$H_a: X \text{ \& } Y \text{ are assoc. } (\beta_1 \neq 0)$

test $\rightarrow \hat{\beta}_1 - 0$ *No assumpt.*

(t-score) $t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)} \sim t_{df}(0, 1)$

df = n - 1 (small, t-dist better)

n > 30 $t_{df} \sim N$



If $p\text{-val} < 0.05$, reject H_0

How to measure accuracy (S.L.R)?

2 key metrics: ① RSE (residual standardized error)

MSE (7) vs $df = n-2$ ② R^2 (percent of variance expl)

(Same units as y)

$$RSE = \sqrt{\frac{1}{n-2} RSS} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}$$

corrected avg of var LR captures

$[0, \infty)$
Best
Perfect
LR.

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

%

$[0, 1]$
Worstest

$$RSS = \sum (y_i - \hat{y}_i)^2, \quad TSS = \sum (y_i - \bar{y})^2$$

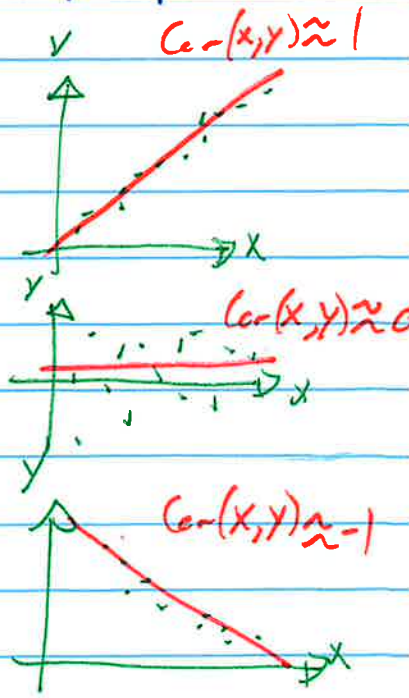
Var(res) ish total Var(Y)

Usually use both, give diff perspectives

In S.L.R., can show

$$R^2 = r^2 = \text{Cor}(X, Y)^2$$

$-1 \leq \text{Cor}(X, Y) \leq 1$
measures direction
fit. $(\hat{\beta}_1!)$



$$Y = \beta X + \epsilon$$

Why

sample

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

"avg" $\bar{x}$ n -samples n data pts

pg 3

stat $s \approx \sigma$?
 ↑ sample ↑ pop

What if

$$s = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2}$$

sample again

$$E[(s)^2] = \dots = \left(\frac{n-1}{n} \right) \sigma^2$$

Almost σ^2



props
of
exp

x_i 's rand
vars

If want $E[(s^2)] = \dots = \sigma^2$

need

$$s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

df = n-1

Multiple Linear Regression (Samirish as)

S.L.R.

$$Y = f(X) + \varepsilon = (\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p) + \varepsilon$$

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{bmatrix}$$

(p indep vars)

$$Y \approx \hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \dots + \hat{\beta}_p X_p$$

 $\hat{\beta}_i$ formulas via minimizing

$$RSS = \sum e_i^2 = \sum (y_i - \hat{y}_i)^2$$

via

① vector calculus $\frac{d}{d\vec{\beta}}$

$$\textcircled{2} \text{ LA } X\vec{\beta} = \vec{y} \text{ no soln}$$

$$X^T X \vec{\beta} = X^T \vec{y} \quad \left(\begin{array}{l} \text{normal} \\ \text{eqns} \end{array} \right)$$

$$\vec{\beta} = (X^T X)^{-1} X^T \vec{y}$$

Inference on $\hat{\beta}$:

$$95\% \text{ CI: } \hat{\beta}_i \pm SE(\hat{\beta}_i)$$

$$\textcircled{H_0} H_0: X \text{ + } Y \text{ not assoc } (\beta_i = 0, i=1, \dots, p)$$

$$H_a: X \text{ + } Y \text{ are assoc } (\text{At least one } \beta_i \neq 0)$$

Cannot use a single dist, use combined distribution call F dist. (ANOVA)

Many normal dist combined.

Sample vs H_0

;)

$$(F\text{-stat}) \quad F = \frac{(TSS - RSS) / p}{RSS / (n - p - 1)}$$

predictors (X_i 's) p

SE isL

F in $[1, \infty)$, $F \gg 1$, H_0 not true

$\uparrow$
 H_0 true

F-stat yields a p-val via F-distr.

Measure accuracy:

$$RSE = \sqrt{\frac{1}{n - p - 1} RSS} \quad [0, \infty)$$

Same as SLR

$$R^2 = \frac{TSS - RSS}{TSS} \quad \left(\begin{array}{c} \% \text{ var} \\ \text{expl} \end{array} \right) \quad [0, 1]$$

Goal: Pitfalls of diagnostics

Recap: Multiple Linear Regression w/ β_i 's

$$Y = f(X) + \varepsilon \stackrel{\text{assume.}}{=} (\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p) + \varepsilon$$

$$\beta_i \approx \hat{\beta}_i \text{ via least squares (formulas!)}$$

$$\text{Conf SE}(\hat{\beta}_i) < \hat{\beta}_i \text{ CI}$$

HT on the entire linear model $f(X)$ via
F-score & F-distr.

Accuracy: Same as simple L.R.

MSE

$$RSE = \sqrt{\frac{1}{n-(p+1)} \text{RSS}} \text{ (y-units)}$$

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \text{ (\% var. expl in } [0,1])$$

CS/ML Avg sq. errors

Pitfalls ① Bad fit (low R^2)

② Sensitive fit (train data small change)
 β_i 's move a lot

③ Incorrect interpretation (of β_i 's)

Dummy
vars

Fix: ① Change model

$\beta_0 + \beta_1 X + \beta_2 X^2$ Hard to interp
 $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2$ Separate Interaction
 $X \rightarrow \ln(X), e^X, \beta_3 Z$

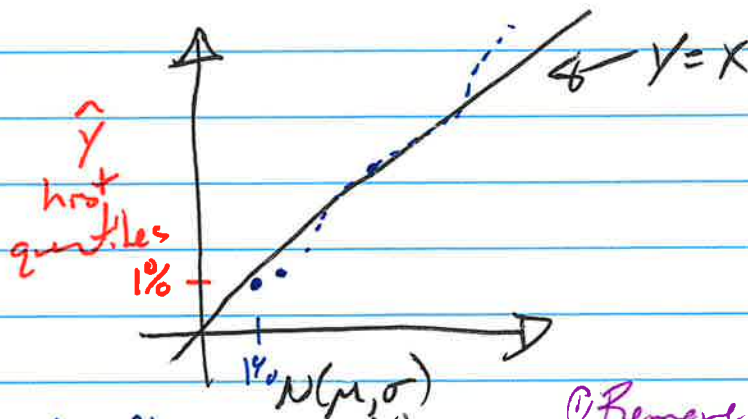
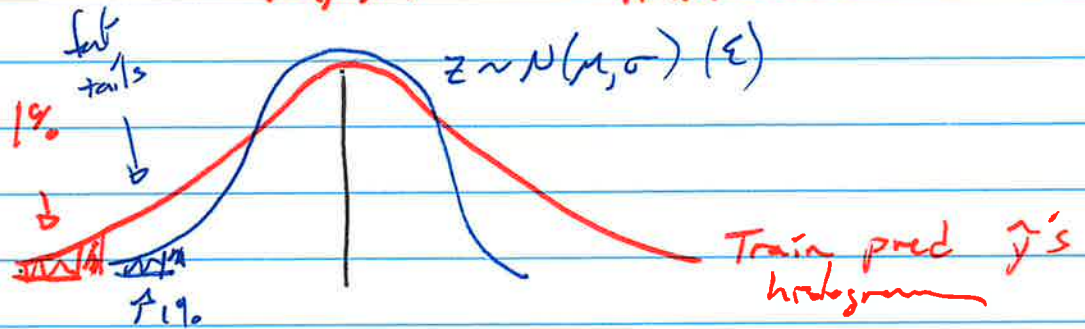
② Change data

↳ data trans

Issue 1 Bad model fit (low R^2)

Linearity assumption not quite right.
How to see? R 4 diag plots (+ R^2)

Z plots on $\hat{y}_i$ vs residuals e_i quantile
I plot on normality of ϵ/Y (Q-Q plot) train data
 $N(\mu, \sigma)$



Issue 2 Sensitive fit quantiles
 leverage
I plot on leverage

① Remove outliers
 ② Change model/trans data

Point X extreme from rest train data.
 Problem if have high error also.

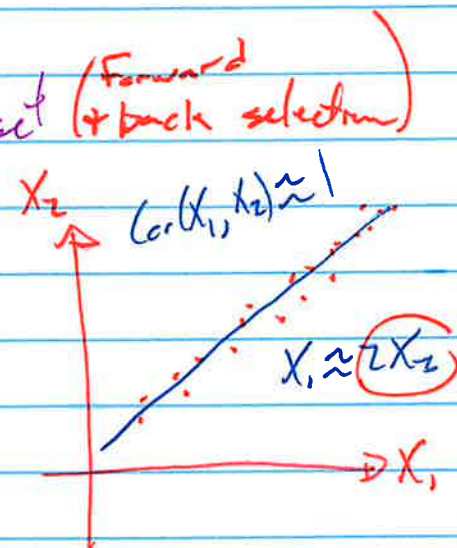
Issue 3 Incorrect interpretation.

Reasons: ① Insignificant variables
↳ Only choose "best" subset (Forward + back selection)
② Correlated variables
not useful

$$Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2$$
$$= \beta_0 + \beta_1 (2X_2) + \beta_2 X_2$$

confuses model

$$= \beta_0 + (2\beta_1 + \beta_2) X_2$$



Check for multicollinearity
via VIF score
variance inflation
factor

2 vars highly corr, →
drop 1 if care
about β_i interp.

VIF ≈ 1 😊
VIF ≈ 5 😐
VIF ≈ 10 ☹️

text for details.